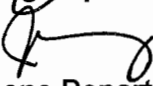


**OPERATOR TRAINING PROGRAM
JOB PERFORMANCE MEASURE**

STATION:	Salem Generating Station		
SYSTEM:	Emergency Operating Procedures		
TASK:	TCAF Total Loss of All Service Water		
TASK NUMBER:	N1140380401		
JPM NUMBER:	15-01 NRC Sim e		
ALTERNATE PATH:	☐	K/A NUMBER:	076 A2.01
APPLICABILITY:		IMPORTANCE FACTOR:	
	EO ☐		3.5 RO
	RO ☒		3.7 SRO
	STA ☐		
	SRO ☒		
EVALUATION SETTING/METHOD:	Simulator / Perform		
REFERENCES:	S2.OP-AB.SW-0001, Loss of SW Header Pressure, Rev. 16 S2.OP-AB.SW-0005, Loss of All Service Water Rev. 4 S2.OP-AR.ZZ-0002, Overhead Annunciator Window B Rev. 36 (All rev checked 8-3-16)		
TOOLS AND EQUIPMENT:	None		
VALIDATED JPM COMPLETION TIME:	5 minutes		
TIME PERIOD IDENTIFIED FOR TIME CRITICAL STEPS:	N/A		
Developed By:	G Gauding Instructor	Date:	3-23-16
Validated By:	W Russell / D Cox SME or Instructor	Date:	8-22-16
Approved By:	 Training Department	Date:	10/11/16
Approved By:	 Operations Department	Date:	10/11/16
ACTUAL JPM COMPLETION TIME:			
ACTUAL TIME CRITICAL COMPLETION TIME:			
PERFORMED BY:			
GRADE:	☐ SAT	☐ UNSAT	
REASON, IF UNSATISFACTORY:			
EVALUATOR'S SIGNATURE:	DATE:		

**OPERATOR TRAINING PROGRAM
JOB PERFORMANCE MEASURE**

NAME: _____

DATE: _____

SYSTEM: Abnormal Operating Procedures

TASK: TCAF a Total Loss of All Service Water

TASK: N1140380401

Simulator Setup: IC-255 100% power. **21 charging pump in service**
2 SW bay removed from service IAW S2.OP-AB.SW-003,
Service Water Bay Leak. All 4 bay pumps in service.
S2.OP-AB.SW-003 out and marked up.

MALFS: Tied to RT-1

24 SW pump trip
25 SW pump trip
26 SW pump trip

INITIAL CONDITIONS:

Unit 2 is operating at 100% power.

21 charging pump is in service.

23 charging pump is aligned for Normal operation IAW Section 4.1 of S2.OP-SO.CVC-0002,
Charging Pump Operation.

#2 SW Bay has been removed from service to isolate a leak in the bay IAW S2.OP-AB.SW-0003,
Service Water Bay Leak.

All #4 Service Water Bay pumps are in service.

INITIATING CUE:

You are the Reactor Operator. Respond to all indications and alarms.

Successful Completion Criteria

1. All critical steps completed.
2. All sequential steps completed in order.
3. All time-critical steps completed within allotted time.
4. JPM completed within validated time. Completion time may exceed the validated time if satisfactory progress is being made

Task Standard for Successful Completion:

1. Trip the Rx
2. Stop all RCPs prior to or within 3 minutes of any RCP bearing temperature reaching 175°F or any other RCP trip criteria being met in S2.OP-AB.RCP-001.
3. Place 23 Charging pump in service.
4. Reduce operating CCW pumps to one.

**OPERATOR TRAINING PROGRAM
JOB PERFORMANCE MEASURE**

NAME: _____

DATE: _____

SYSTEM: Abnormal Operating Procedures

TASK: TCAF a Loss of all Service Water

*	STEP No.	STEP	STANDARD	EVAL S/U	COMMENTS
		Simulator Operator: Insert RT-1 after operator assumes watch. This trips all 4 Bay SW pumps.			
			Recognizes that all SW pumps have tripped.		
			Enters S2.OP-AB.SW-0005, Loss of All Service Water directly, or responds IAW ARP for low service water header pressures. ARP steps are after AB.SW-5 steps here in JPM. AB.SW-1 steps are after ARP steps here in JPM.		
*	AB.SW-5 3.1	TRIP the reactor, AND CONFIRM the Reactor Trip.	Trips the Rx using either trip handle, and confirms the Rx trip.		
*	AB.SW-5 3.2	STOP all RCPs.	Depresses stop PBs for all RCPs.		
		GO TO 2-EOP-TRIP-1, Reactor Trip or Safety Injection, AND CONTINUE with this procedure.	Cue: You are to continue with S2.OP-AB.SW-0005 while other operators perform TRIP-1. Evaluator: Silence OHA's as not associated with loss of SW.		
*	AB.SW-5 3.4	CLOSE the following valves to isolate letdown: • 2CV3, 45 GPM ORIFACE	Shuts the open orifice isolation valve and		

**OPERATOR TRAINING PROGRAM
JOB PERFORMANCE MEASURE**

*	STEP No.	STEP	STANDARD	EVAL S/U	COMMENTS
		• 2CV4, 75 GPM ORIFICE • 2CV5, 75 GPM ORIFICE	ensures the remaining orifice isolation valves are shut.		
	AB.SW-5 3.5	INITIATE Attachment 1, Continuous Action Summary.	Initiates Attachment 1, Continuous Action Summary.		
* * *	AB.SW-5 3.6	IF a Centrifugal Charging Pump is in service <u>AND</u> the 23 Charging Pump is <u>IMMEDIATELY</u> available to be placed in service, <u>THEN</u> PERFORM the following : A. VERIFY <u>ALL</u> 21-24 RCP Seal Inlet Temperatures <225°F as indicated on 2CC1 or the Plant Computer B. ENSURE 23 Charging Pump is aligned for Normal Operation IAW S2.OP-SO.CVC-0002(Q) Charging Pump Operation. C. PLACE 23 Charging Pump in MANUAL. D. PLACE Charging Master Flow Controller in MANUAL. E. SET 23 Charging Pump Speed Demand to 10-12%. F. START 23 Charging Pump <u>AND</u> immediately INCREASE Speed Demand to approximately 20% to couple pump. G. ADJUST 23 Charging Pump Speed Demand to obtain desired flow. H. ENSURE Seal Injection Flow 6-12 gpm to each Reactor Coolant Pump not to exceed 40 gpm total Seal	Determines a centrifugal charging pump is in service and 23 charging pump is immediately available to be placed in service.		

**OPERATOR TRAINING PROGRAM
JOB PERFORMANCE MEASURE**

*	STEP No.	STEP	STANDARD	EVAL S/U	COMMENTS
*		Injection Flow. I. STOP 21 and 22 Centrifugal Charging Pumps. J. DISPATCH an Operator to monitor operation of 23 Charging Pump due to the potential for rising Component Cooling Water System temperature.			
	AB.SW-5 3.7	<u>IF</u> a Centrifugal Charging Pump is in service <u>AND</u> the 23 Charging Pump is <u>NOT IMMEDIATELY</u> available to be placed in service, <u>THEN STOP</u> 21 and 22 Centrifugal Charging Pumps.	Determines a Centrifugal Charging Pump is not in service		
	AB.SW-5 3.8	PLACE 2CC131, RCP THERMAL BARRIER, in MANUAL.	Places 2CC131, RCP THERMAL BARRIER, in MANUAL.		
*	AB.SW-5 3.9	PLACE Component Cooling Pumps in MANUAL <u>AND</u> REDUCE the number of operating Component Cooling Water Pumps to one.	Places all Component Cooling Pumps in MANUAL <u>AND</u> stops all but one Component Cooling Water Pumps.		
			Terminate JPM.		
	ARP OHA's B-13,14	3.1 VERIFY header low pressure using 2PA5373 on 2CC1. 3.2 <u>IF</u> low pressure conditions exists due to known system configuration, <u>THEN GO TO</u> S2.OP-SO.SW-0001(Q) Service water Pump operation, to adjust system pressure. 3.3 <u>IF</u> a valid low pressure condition exists, <u>THEN GO TO</u> S2.OP-AB.SW-0001(Q),	Determines header pressure is low. Determines loss of SW system pressure does not conditions exists due to known system configuration. Determines valid low pressure condition exists, and goes to S2.OP-AB.SW-0001, Loss of		

**OPERATOR TRAINING PROGRAM
JOB PERFORMANCE MEASURE**

*	STEP No.	STEP	STANDARD	EVAL S/U	COMMENTS
		Loss of Service Water Header Pressure.	Service Water Header Pressure.		
	AB.SW-1 3.1	INITIATE Attachment 1, Continuous Action Summary.	Initiates Attachment 1, Continuous Action Summary.		
	AB.SW-1 3.2	Is there indication of a leak in a Service Water Bay per the following? • Visual indication • OHA B-29, 21-23 SW PMP SMP AREA LVL HI • OHA B-30, 24-26 SW PMP SMP AREA LVL HI	Determines there is no unisolated leak (SW Bay 2 has been isolated)		
	AB.SW-1 3.4	START Service Water pumps to maintain header pressure between 95 and 105 psig.	Determines no SW pumps are available to be started.		
	AB.SW-1 3.5	Is there a leak or valve malfunction on the turbine header as indicated by any of the following?	Determines the cause of low SW header pressure is from loss of all SW pumps.		
	AB.SW-1 3.6	Are both nuclear Service Water headers in service?	Determines neither nuclear SW header is in service.		
	AB.SW-1 3.7	<u>IF</u> the Unit is in a Service Water Header Outage...	Determines the Unit is not in a Service Water Header Outage.		
	AB.SW-1 3.8	<u>IF</u> the Unit is <u>NOT</u> in a Service Water Header Outage, <u>THEN GO TO</u> S2.OP-AB.SW-0005 (Q), Loss of All Service Water	Determines the Unit is not in a Service Water Header Outage, and goes to S2.OP-AB,SW-0005.		

TERMINATING CUE: After reduction of CCW pumps to one, terminate JPM.

OPERATOR TRAINING PROGRAM
JOB PERFORMANCE MEASURE

TQ-AA-106-0303
Revision 5

JOB PERFORMANCE MEASURE VALIDATION CHECKLIST

NOTE: All steps of this checklist should be performed upon initial validation. Prior to JPM usage, revalidate JPM using steps 8 and 11 below.

- W ✓ W ✓ 1. Task description and number, JPM description and number are identified.
- W ✓ W ✓ 2. Knowledge and Abilities (K/A) references are included.
- W ✓ W ✓ 3. Performance location specified. (in-plant, control room, or simulator)
- W ✓ W ✓ 4. Initial setup conditions are identified.
- W ✓ W ✓ 5. Initiating and terminating Cues are properly identified.
- W ✓ W ✓ 6. Task standards identified and verified by SME review.
- W ✓ W ✓ 7. Critical steps meet the criteria for critical steps and are identified with an asterisk (*).
- W ✓ W ✓ 8. Verify the procedure referenced by this JPM matches the most current revision of that procedure: Procedure Rev. 4 Date 8/22/10
- W ✓ W ✓ 9. Pilot test the JPM:
 - a. verify Cues both verbal and visual are free of conflict, and
 - b. ensure performance time is accurate.
- _____ 10. If the JPM cannot be performed as written with proper responses, then revise the JPM.
- _____ 11. When JPM is revalidated, SME or Instructor sign and date JPM cover page.

SME/Instructor: W. Russell Russell

Date: 8/22/16

SME/Instructor: D. Cox

Date: 8-22-16

SME/Instructor: _____

Date: _____

**OPERATOR TRAINING PROGRAM
JOB PERFORMANCE MEASURE**

INITIAL CONDITIONS:

Unit 2 is operating at 100% power.

21 charging pump is in service.

23 charging pump is aligned for Normal operation IAW Section 4.1 of S2.OP-SO.CVC-0002,
Charging Pump Operation.

#2 SW Bay has been removed from service to isolate a leak in the bay IAW S2.OP-AB.SW-0003,
Service Water Bay Leak.

All #4 Service Water Bay pumps are in service.

INITIATING CUE:

You are the Reactor Operator. Respond to all indications and alarms.